Peter **Asanov**

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About

Senior software engineer with 20+ years of experience, including over a decade building real-time, data-intensive, and mission-critical distributed systems. Builds and owns cloud, on-premises, and edge platforms from architecture to production, with a strong focus on security. Contributor to widely used open-source projects.

Skills

Distributed Systems — HA/fault tolerance, scalability, partitioning/replication, EDA and messaging, consensus (Raft, Paxos), CAP trade-offs/consistency models

Development

C++11–23 — Boost/Asio, Cap'n Proto, FlatBuffers, Seastar
Rust — Tokio, hyper, PyO3/maturin
Go — Gin, zap/slog
Python — FastAPI/Pydantic, SQLAlchemy, Celery, NumPy, pandas, Polars
JVM — Java/Scala/Kotlin, Spring Boot/Hibernate, Akka
JS/TS — React, Vue, Vite/Vitest, tRPC, Prisma
Build — Nix, CMake/Ninja, Conan, Maven, Bazel, Meson

Computer Systems

CPU/concurrency — x86-64/ARM64 assembly, SIMD (SSE/NEON), memory models, memory and instruction ordering, fences/barriers, multithreading/multiprocessing
HW-aware optimization
Linux — task scheduling, VM/paging, seccomp/SELinux, io_uring
Networking — TCP/IP, UDP, DNS, routing/NAT, QoS, L4/L7 load balancing, virtual/overlay networks (Tailscale), eBPF/XDP/AF_XDP, Unix/raw sockets
Virtualization/containers — Proxmox, QEMU/KVM, Cloud Hypervisor, LXC/Incus, Docker Engine/containerd

Platform & Infrastructure

Bare-metal/edge — iPXE, Matchbox (Terraform provider), Flatcar/systemd-sysext, Keylime (TPM 2.0 attestation)
Cloud — AWS (EC2/ECS/Lambda), Azure (VM/Functions), GCP (GCE/App Engine/Functions)
Kubernetes — Talos Linux, k3s, EKS, AKS, GKE, OpenShift, Helm, Kustomize
Gateway API, Cilium (CNI), kube-vip, MetalLB
Strimzi, CloudNativePG, cert-manager, Trivy Operator, Prometheus Operator, kube-state-metrics
IaC — Terraform, Tofu Controller, Ansible, CDK, CloudFormation, Bicep/ARM templates, NixOps4
CI/CD & GitOps — Flux CD, Argo CD, GitHub Actions, GitLab CI/CD, Azure DevOps
Observability — Prometheus, Grafana/LGTM, OTel/OTLP, CloudWatch/X-Ray, Azure Monitor/App Insights

Security

Security design — PoLP, authz (ABAC, RBAC), trust models and Zero Trust, attestation, isolation, audit
IAM/IdPs — OAuth 2.0/OIDC (PKCE/mTLS/DPoP), SAML, Entra ID/Azure AD, Cognito, Auth0
Cryptography — Curve25519, ECDH, EdDSA, Ristretto, ZK-SNARKs/ZK-STARKs

Data

DBs — Postgres (PostGIS, TimescaleDB), DynamoDB, MySQL, Cassandra, MongoDB, DuckDB

KV/cache	Redis, etcd, RocksDB, Valkey
Object storage	S3, MinIO, RustFS, Azure Blob, GCS
Messaging	Kafka, RabbitMQ, Pulsar
Pipelines	Airflow, Spark/PySpark, Flink, NiFi

AI/ML

NLP	POS tagging, lemmatization, dependency parsing, NER, NLTK, spaCy, Stanza
LLM integration	RAG, tool integration (MCP), API interoperability

FOSS contributions



Tailscale

tailscale/tailscale

Zero Trust identity-based connectivity platform

- Unix-domain socket support for Tailscale Serve, exposing local HTTP services over the tailnet without a TCP bridge.



ipsw

blacktop/ipsw

iOS/macOS Research Swiss Army Knife

- dyld shared-cache fixes: address mapping, v4 rebasing and v5 pointer decoding.
- Objective-C metadata parsing and ExclaveKit runtime resolution.
- C++ symbol demangling for compiler-outlined cold functions.

Work Experience

Senior Software Engineer | Platform Architecture & Security

Netherlands

Nebul

Started in Aug 2026

Own the architecture, planning and implementation of a declarative systems management platform for Nebul's datacenter infrastructure, designed for isolated networks and combining configuration enforcement, endpoint security and compliance reporting.

Contributions:

- Consolidate over 30 ADRs into a coherent system specification by building NLP-assisted pipeline for architectural review and decision traceability.
- Unify and extend existing security controls into a system-wide model, defining trust relationships and least-privilege enforcement at service and operating-system levels. Validate fail-closed process supervision through crash-injection tests. Design integration with Nebul's node provisioning platform to anchor agent identity in each node's TPM 2.0 root of trust.
- Design operational logging and audit flows, including durable central collection of node audit records and verified server acknowledgement of operation intent before sensitive changes.
- Establish the engineering foundation: reproducible builds, CI and dev workflow.
- Prototype agent communication for isolated networks using outbound-only connections (HTTP/3, mTLS).

Tech: Go, Protobuf Editions/Buf/gRPC; Nix/flake-parts, GitLab CI, Talos Linux, Argo CD; mTLS, QUIC/WebTransport, HTTP/2 CONNECT; OpenBao (PKI, Transit), cosign, TPM 2.0; SELinux, bubblewrap, seccomp, Linux namespaces, cgroup v2; Python/spaCy, NetworkX, Markdown AST (mdast)

Senior Software Engineer

Rotterdam, Netherlands

The Ocean Cleanup

Mar 2026 – Jul 2026

Owned, re-architected, and shipped the infrastructure for Onboard Compute, an edge platform for UAV and vessel data ingestion, visualization, operational monitoring, and offshore cleanup planning.

- Scaled k3s from one node to a three-server HA cluster with an embedded etcd quorum; added Kubernetes API endpoint failover and an ingress VIP reachable from both the onboard LAN and tailnet.
- Introduced GitOps for in-cluster state and Ansible for node provisioning, access management, and k3s/Flux bootstrap. Secured runtime secrets with SOPS; validated fully rendered Flux trees with KubeLinter and strict kubeconform checks against pinned schemas; and built a topology-equivalent Tart/k3s environment for validation without field-cluster access.

- Authored an internal RFC to standardize Gateway API routing across edge and cloud systems and developed a Talos Linux PoC for immutable, API-managed edge nodes.
- Improved map legibility with higher-contrast density bands and clearer hierarchy between vessel and route layers.
- Instrumented UAV ingestion for Zenoh freshness and forwarding, Redis Streams/DLQ processing, and end-to-end latency; added alerts for Flux reconciliation, ingress health, and remote-write failures.
- Enabled unmodified amd64 workloads on Apple Silicon by replacing k3s's embedded containerd with Docker Engine and registering Rosetta through binfmt_misc, bypassing an OCI runtime/seccomp incompatibility.

Platform: k3s, Flux CD, kube-vip, MetalLB (L2), Tailscale, Traefik, CloudNativePG, PostgreSQL (PostGIS, TimescaleDB), Redis Streams, Zenoh, Prometheus/Grafana, RustFS

Tooling & app: Ansible, Kustomize, SOPS, KubeLinter, kubeconform, Docker Engine, Tart/Rosetta, Nix/flake-parts, Python, TypeScript/Vue

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Senior Software Engineer | Partner Integrations — Amsterdam, Netherlands

Vox AI — Jul 2025 – Oct 2025

Shipped backend integrations for a voice-AI drive-thru platform, working directly with founders.

- Delivered an OAuth 2.0/OIDC integration bridging AWS Cognito and a third-party DSP protocol within two weeks, enforcing PKCE while handling non-compliant partner clients.
- Built a multi-tenant location-management API on DynamoDB; added correlation IDs, X-Ray, Lambda Powertools, and EventBridge/SQS monitoring across authorization, token, and webhook flows.
- Structured the CDK stack to avoid 15–40 minutes of Cognito/CloudFront reprovisioning per iteration.
- Built a deterministic POS-to-RAG transformation pipeline that resolved ambiguity, circular dependencies, and orphaned references while remaining stable across upstream schema changes; integrated data-quality checks into CI.

Platform: AWS Lambda, API Gateway, Cognito, DynamoDB, X-Ray, CloudWatch, EventBridge, SQS, CDK

App stack: TypeScript, Python 3.12+, Pydantic, FastAPI, Mangum, Moto, pytest/pytest-asyncio, Vitest, HTTPX, uv

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Senior Software Engineer — Netherlands

Selected contracts — 2022 – Present

- **Deliverect:** extended the data model and API behavior of a restaurant-automation PaaS.
  Stack: Python, Eve, Celery, GKE, TypeScript/React, Redis, MinIO, Terraform, Helm
- **Ahold Delhaize (Albert Heijn):** built an online data ingestion platform for internal compliance and R&D analytics.
  Stack: Go, AKS, Strimzi/Kafka, Argo CD, Airflow, Spark, Terraform, Helm
- **Private client:** built taxi-terminal telemetry ingestion for compliance and fraud analysis.
  Stack: Go, AKS, Kafka, PostgreSQL, Terraform, Helm
- **LEGO:** built internal IAM and identity-provider functionality.
  Stack: Azure Functions, SAML IdP, Python
- **Private client:** built a cloud video-conversion service.
  Stack: C++20, CMake/Ninja, Go, gRPC-Gateway, FFmpeg, EKS, CloudWatch, Airflow, Nix, Terranix, Helm
- **Private client:** built a custom video-calling service.
  Stack: C++23, CMake/Ninja, Asio, libdatachannel (WebRTC), Nix, Ansible

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Core Software Engineer — Haarlem, Netherlands

AIMMS B.V. — Nov 2019 – Sep 2021

Built core platform and migration tooling as a member of the AIMMS PRO core team:

- Migrated Java and Scala backend services from ECS to AKS; configured landing-zone RBAC, bootstrapped Terraform state, and delivered supporting IaC.
- Built a C++17 microservice template with an API-first protobuf DSL for gRPC and REST surfaces; contributed to Conan/Artifactory developer-platform infrastructure; built PoCs and delivered demos for the tooling.
- Improved cost efficiency and throughput predictability for compute-heavy, nondeterministic optimization workloads by implementing Kubernetes-native bin packing with resource requests/limits, affinity, priority classes, pre-warmed capacity, Cluster Autoscaler, and taints/tolerations.

- Built migration tooling for sensitive, business-critical customer data in Amazon RDS and a declarative-schema CRUD generator for administrative and internal tools.

Stack: ECS, EC2, RDS, CloudWatch, Go, Python, Java, Scala, C++17, TypeScript, React, Conan, Artifactory, CMake, Ninja, gRPC-Gateway, Kubernetes (AKS), Application Insights, Terraform, Helm

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Senior Software Engineer — Amsterdam, Netherlands

HVR Software (acquired by Fivetran) — Jan 2018 – Oct 2019

- Worked on HVR's high-throughput, log-based replication engine. Reverse-engineered Oracle histogram internals to enable a specific HVR sharding use case.
- Rewrote the feature-flag DSL-to-C transpiler used in every HVR build, including multiple-inheritance and range-override resolution. Achieved an order-of-magnitude speedup, substantially reducing CI compute costs. Built an HTML documentation generator, extending the DSL with annotation support.
- Built a Go REST service for cloud integration.

Stack: C99, Python, Go, RDBMS and big-data integrations, Oracle (histogram internals), Airflow, Spark

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Software Engineer — Eindhoven, Netherlands

Mapscape B.V. — 2017

Implemented backend components of the Mapscape NDS compiler for raw map data parsing and introduced parallel parsing to reduce processing time.

Stack: C++11/14, Boost, GTest, SQLite, shell, Python, Airflow

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Senior Software Engineer — St. Petersburg, Russia

New Cloud Technologies — 2014 – 2017

Built collaboration-engine features for an on-premises office suite and developed its spreadsheet client for Samsung Tizen.

- Independently built a cross-platform FreeType-based font-rendering engine with subpixel positioning for scalable document layouts and integrated it into a large enterprise codebase.
- Developed advanced spreadsheet editing for Samsung Tizen using a declarative, compile-time state machine to avoid dynamic dispatch overhead and resolve editing edge cases observed in competing web suites.

Stack: C++14, CMake, Emscripten, Objective-C, Qt, SWIG, Kubernetes

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Senior Software Engineer — St. Petersburg, Russia

AIS — 2014

Built C++11 microservices for the banking and exchange subsystem of a cryptocurrency mining pool:

- Designed an Asio-based service framework and implemented all C++11 banking services, including multi-stage, fault-tolerant transactions.
- Protected financial state with append-only storage, two-factor transactions, circuit breakers, alerting, rollback paths, and extensive testing.
- Shipped the zero-to-one production release in under two months with a three-engineer team, owning the C++ subsystem end to end.

Stack: C++11, Core Java, Groovy, Ruby (Chef)

Education

B.Sc. in Computer Science and Engineering — Novosibirsk, Russia

Novosibirsk State Technical University (NSTU) — 2002 – 2007

Previous work history

Available upon request — 2005 – 2014